

The factorial function $f: \mathbb{N} \rightarrow \mathbb{Z}^+$, denoted by $f(n) = n!$. The value of $f(n) = n!$ is the product of the first n positive integers, so $f(n) = 1 \cdot 2 \cdot \dots \cdot (n-1) \cdot n$ [and $f(0) = 0! = 1$].

Example 33: $f(1) = 1! = 1$, $f(2) = 2! = 1 \cdot 2 = 2$, and $f(6) = 6! = 720$

$$f(20) = 20! = 2,432,902,008,176,640,000 \cdot \square$$

Stirling's approximation: $n! \sim \sqrt{2\pi n} (n/e)^n$

2.3.6. Partial Functions

Definition 13: A partial function f from a set A to a set B is an assignment to each element a in a subset of A , called the domain of definition of f , of a unique element b in B . The sets A and B are called the domain and codomain of f , respectively. We say that f is undefined for elements in A that are not in the domain of definition of f . When the domain of defn. of f equals A , we say that f is a total function.

Example 34: The function $f: \mathbb{Z} \rightarrow \mathbb{R}$ where $f(n) = \sqrt{n}$ is a partial function from $\mathbb{Z}$ to $\mathbb{R}$ where the domain of definition is the set of nonnegative integers. Note that f is undefined for negative integers. $\square$